MPG No. 7.3.22 $m'm2'$ (m'm2' setting) [Type III, orthorhombic] [T tensor]

* Rank 0 tensor. * Rank 1 tensor.

$$\begin{bmatrix} T_x & 0 & 0 \end{bmatrix}$$

$$T_x = \mathbb{T}_x^{(1)}$$

* Rank 2 tensor (s).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = \mathbb{T}_{xz}^{(1)}$$

* Rank 2 tensor (a).

$$\begin{bmatrix} 0 & 0 & T_{xz} \\ 0 & 0 & 0 \\ -T_{xz} & 0 & 0 \end{bmatrix}$$

$$T_{xz} = -\mathbb{M}_y^{(1)}$$

* Rank 3 tensor (ss).

$$\begin{bmatrix} T_{xxx} & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & T_{zzx} \\ 0 & T_{yyx} & 0 \end{bmatrix}$$

$$T_{xxx} = 3\mathbb{T}_x^{(1)} + 2\mathbb{T}_x^{\alpha(1)}$$
$$T_{yyx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} + \mathbb{T}_x^{\beta(1)}$$
$$T_{zzx} = \mathbb{T}_x^{(1)} - \mathbb{T}_x^{\alpha(1)} - \mathbb{T}_x^{\beta(1)}$$

* Rank 3 tensor (sa).

$$\begin{bmatrix} 0 & 0 & 0 \\ T_{yyx} & 0 & 0 \\ T_{zzx} & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\frac{T_{zzx}}{2} \\ 0 & -\frac{T_{yyx}}{2} & 0 \end{bmatrix}$$

$$T_{yyx} = 2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$
$$T_{zzx} = -2\mathbb{M}_{yz}^{(1)} + 2\mathbb{T}_x^{(2)}$$
$$\frac{T_{zzx}}{2} = -\mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$
$$\frac{T_{yyx}}{2} = \mathbb{M}_{yz}^{(1)} + \mathbb{T}_x^{(2)}$$

* Rank 3 tensor (a).

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & T_{zxz} \\ 0 & T_{xyy} & 0 \end{bmatrix}$$

$$T_{zxz} = \mathbb{M}^{(2)}_{yz} + \mathbb{T}^{(3)}_{x}$$
$$T_{xyy} = \mathbb{M}^{(2)}_{yz} - \mathbb{T}^{(3)}_{x}$$

* Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ T_{xxzx} & T_{yyzx} & T_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = -\mathbb{T}^{\alpha(1)}_{4y} - \mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{xz}$$
$$T_{yyzx} = 2\mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz}$$
$$T_{zzzx} = \mathbb{T}^{\alpha(1)}_{4y} - \mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(1)}_{xz} + 2\mathbb{T}^{(2)}_{xz}$$
$$T_{yzxy} = 2\mathbb{T}^{\beta(1)}_{4y} + \mathbb{T}^{(2)}_{xz}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & 0 & T_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & T_{yyzx} & 0 \\ 0 & 0 & 0 & 0 & T_{zzzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & T_{yzxy} \\ -T_{xxzx} & -T_{yyzx} & -T_{zzzx} & 0 & 0 & 0 \\ 0 & 0 & 0 & -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{xxzx} = -2\mathbb{M}^{(1)}_{y} + 2\mathbb{M}^{\alpha(1)}_{y} + 2\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz}$$
$$T_{yyzx} = -4\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz}$$
$$T_{zzzx} = 2\mathbb{M}^{(1)}_{y} - 2\mathbb{M}^{\alpha(1)}_{y} + 2\mathbb{M}^{\beta(1)}_{y} + \mathbb{T}^{(3)}_{xz}$$
$$T_{yzxy} = \mathbb{M}^{(1)}_{y} + 4\mathbb{M}^{\alpha(1)}_{y}$$

* Rank 4 tensor (aas).

$$\begin{bmatrix} 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = 2\mathbb{T}^{(4)}_{xz}$$

* Rank 4 tensor (aaa).

$$\begin{bmatrix} 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ -T_{yzxy} & 0 & 0 \end{bmatrix}$$

$$T_{yzxy} = \mathbb{M}^{(2)}_{y}$$

* Rank 4 tensor (sa).

$$\begin{bmatrix} 0 & T_{xxzx} & 0 \\ 0 & T_{yyzx} & 0 \\ 0 & T_{zzzx} & 0 \\ 0 & 0 & T_{yzxy} \\ 0 & 0 & 0 \\ T_{xyyz} & 0 & 0 \end{bmatrix}$$

$$
\begin{aligned}
T_{xxzx} &= \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{xz}^{(5)} \\
T_{yyzx} &= \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} \\
T_{zzzx} &= \mathbb{M}_y^{(3)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{yzxy} &= \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{xz}^{(5)} \\
T_{xyyz} &= \mathbb{M}_y^{(4)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{xz}^{(5)}
\end{aligned}
$$

* Rank 4 tensor (as).

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & T_{yzxy} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & 0 & 0 \\
0 & 0 & 0 & T_{xyyz} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxy} &= \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} - \mathbb{T}_{xz}^{(6)} \\
T_{zxxx} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} - \mathbb{M}_y^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyy} &= \mathbb{M}_y^{(5)} + 2\mathbb{M}_y^{(6)} + 2\mathbb{M}_y^{\alpha(3)} \\
T_{zxzz} &= \mathbb{M}_y^{(5)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\
T_{xyyz} &= \mathbb{M}_y^{(6)} - \mathbb{M}_y^{\alpha(3)} + \mathbb{M}_y^{\beta(3)} + \mathbb{T}_{xz}^{(6)}
\end{aligned}
$$

* Rank 4 tensor (s).

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & T_{xxzx} & 0 & 0 & T_{xxxz} & 0 \\
0 & 0 & 0 & 0 & T_{yyzx} & 0 & 0 & T_{yyxz} & 0 \\
0 & 0 & 0 & 0 & T_{zzzx} & 0 & 0 & T_{zzxz} & 0 \\
0 & 0 & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzyx} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & T_{xyyz} & 0 & 0 & T_{xyzy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxzx} &= -2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} \\
T_{xxxz} &= -2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{yyzx} &= \mathbb{M}_y^{(3)} + 2\mathbb{M}_y^{(4)} + 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} \\
T_{yyxz} &= -\mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{(4)} - 2\mathbb{M}_y^{\alpha(2)} - 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} \\
T_{zzzx} &= 2\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} + \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{zzxz} &= 2\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(3)} - 2\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + 2\mathbb{M}_y^{\beta(1)} - \mathbb{M}_y^{\beta(2)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} \\
T_{yzxy} &= \mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)} \\
T_{yzyx} &= \mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} + 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} \\
T_{zxxx} &= 2\mathbb{M}_y^{(1)} - 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} - \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} \\
T_{zxyy} &= 4\mathbb{M}_y^{\beta(1)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} - \mathbb{T}_{xz}^{(3)} \\
T_{zxzz} &= -2\mathbb{M}_y^{(1)} + 2\mathbb{M}_y^{\alpha(1)} - 2\mathbb{M}_y^{\beta(1)} + \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} \\
T_{xyyz} &= -\mathbb{M}_y^{(1)} + \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} - \mathbb{M}_y^{\alpha(2)} - \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)} \\
T_{xyzy} &= -\mathbb{M}_y^{(1)} - \mathbb{M}_y^{(4)} - 4\mathbb{M}_y^{\alpha(1)} + \mathbb{M}_y^{\alpha(2)} + \mathbb{M}_y^{\beta(2)} + 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(5)}
\end{aligned}
$$

* Rank 4 tensor (a).

$$
\begin{bmatrix}
0 & 0 & 0 & 0 & 0 & T_{yzxy} & 0 & 0 & T_{yzyx} \\
T_{zxxx} & T_{zxyy} & T_{zxzz} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & T_{xyyz} & 0 & 0 & T_{xyzy} & 0 & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{yzxy} &= \mathbb{M}_{y}^{(2)} + \mathbb{M}_{y}^{(6)} - \mathbb{M}_{y}^{\alpha(3)} - \mathbb{M}_{y}^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\
T_{yzyx} &= -\mathbb{M}_{y}^{(2)} + \mathbb{M}_{y}^{(6)} - \mathbb{M}_{y}^{\alpha(3)} - \mathbb{M}_{y}^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} - \mathbb{T}_{xz}^{(6)} \\
T_{zxxx} &= \mathbb{M}_{y}^{(5)} - \mathbb{M}_{y}^{\alpha(3)} - \mathbb{M}_{y}^{\beta(3)} + 2\mathbb{T}_{xz}^{(6)} \\
T_{zxyy} &= \mathbb{M}_{y}^{(5)} + 2\mathbb{M}_{y}^{(6)} + 2\mathbb{M}_{y}^{\alpha(3)} \\
T_{zxzz} &= \mathbb{M}_{y}^{(5)} - \mathbb{M}_{y}^{\alpha(3)} + \mathbb{M}_{y}^{\beta(3)} - 2\mathbb{T}_{xz}^{(6)} \\
T_{xyyz} &= -\mathbb{M}_{y}^{(2)} + \mathbb{M}_{y}^{(6)} - \mathbb{M}_{y}^{\alpha(3)} + \mathbb{M}_{y}^{\beta(3)} + 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)} \\
T_{xyzy} &= \mathbb{M}_{y}^{(2)} + \mathbb{M}_{y}^{(6)} - \mathbb{M}_{y}^{\alpha(3)} + \mathbb{M}_{y}^{\beta(3)} - 2\mathbb{T}_{xz}^{(4)} + \mathbb{T}_{xz}^{(6)}
\end{aligned}
$$

* Rank 4 tensor (t).

$$
\begin{bmatrix}
0 & 0 & T_{xxxz} \\
0 & 0 & 0 \\
T_{zzzx} & 0 & 0 \\
T_{yyzx} & 0 & 0 \\
0 & 0 & T_{zzxz} \\
0 & 0 & 0 \\
0 & 0 & 0 \\
T_{zxxx} & 0 & 0 \\
0 & 0 & T_{xyyz} \\
0 & T_{xyyz} + \frac{T_{zzxz}}{2} - \frac{T_{zzzx}}{2} & 0
\end{bmatrix}
$$

$$
\begin{aligned}
T_{xxxz} &= -\mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{zzzx} &= \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} - 2\mathbb{T}_{xz}^{(5)} \\
T_{yyzx} &= 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + \mathbb{T}_{xz}^{(3)} \\
T_{zzxz} &= \mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} + \mathbb{T}_{xz}^{(3)} + 2\mathbb{T}_{xz}^{(5)} \\
T_{zxxx} &= -\mathbb{T}_{4y}^{\alpha(1)} - \mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(1)} + 2\mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(3)} \\
T_{xyyz} &= 2\mathbb{T}_{4y}^{\beta(1)} + \mathbb{T}_{xz}^{(2)} - \mathbb{T}_{xz}^{(5)}
\end{aligned}
$$